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# STEP 1: IMPORT LIBRARIES
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.linear_model import LinearRegression



# STEP 2: LOAD DATASET
from google.colab import files
uploaded = files.upload()

data = pd.read_csv('headbrain.csv')

# Display first few rows
print("Dataset Preview:\n", data.head())



# STEP 3: PREPARE DATA
# Extract independent (X) and dependent (Y)
x = np.array(data['Head Size(cm^3)'])
y = np.array(data['Brain Weight(grams)'])

print("\nFirst 5 values of X:", x[:5])
print("First 5 values of Y:", y[:5])



# STEP 4: MANUAL LINEAR REGRESSION
def get_line(x, y):
    # Calculate mean
    x_mean = np.mean(x)
    y_mean = np.mean(y)

    print("\nMean of X:", x_mean)
    print("Mean of Y:", y_mean)

    # Calculate slope (m) and intercept (c)
    x_diff = x - x_mean
    y_diff = y - y_mean

    m = np.sum(x_diff * y_diff) / np.sum(x_diff ** 2)
    c = y_mean - (m * x_mean)

    print("Slope (m):", m)
    print("Intercept (c):", c)

    # Return equation y = mx + c
    return lambda val: m * val + c


# Create regression function
line = get_line(x, y)


# STEP 5: PLOT REGRESSION LINE
X = np.linspace(np.min(x) - 100, np.max(x) + 100, 1000)
Y = np.array([line(val) for val in X])

plt.plot(X, Y, color='red', label='Regression Line')
plt.scatter(x, y, color='green', label='Data Points')

plt.xlabel('Head Size (cm^3)')
plt.ylabel('Brain Weight (grams)')
plt.title('Manual Linear Regression')
plt.legend()
plt.show()


# STEP 6: CALCULATE R² SCORE (MANUAL)
def get_error(line_func, x, y):
    y_mean = np.mean(y)
    y_pred = np.array([line_func(val) for val in x])
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    ss_total = np.sum((y - y_mean) ** 2)
    ss_residual = np.sum((y - y_pred) ** 2)

    r2 = 1 - (ss_residual / ss_total)
    return r2

print("\nR² Score (Manual):", get_error(line, x, y))


# STEP 7: USING SKLEARN LIBRARY
# Reshape X for sklearn
x_reshaped = x.reshape(-1, 1)

# Create model
model = LinearRegression()

# Train model
model.fit(x_reshaped, y)

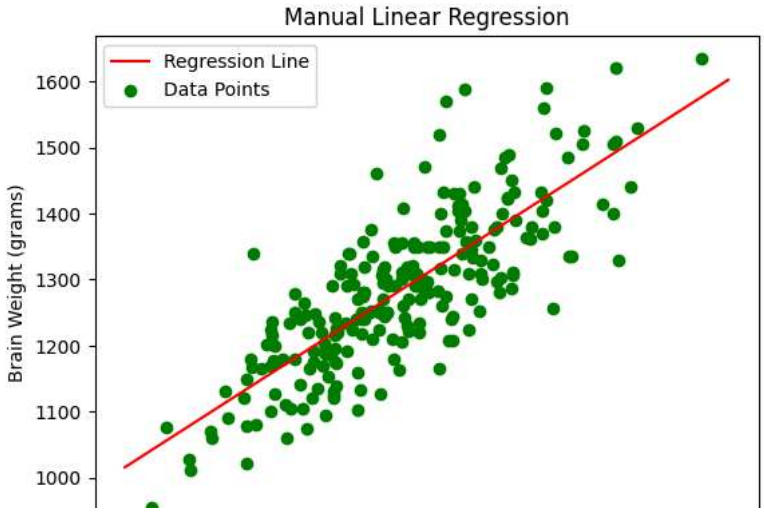
# Predict values
y_pred = model.predict(x_reshaped)

# Print results
print("\nSklearn Results:")
print("Coefficient (m):", model.coef_)
print("Intercept (c):", model.intercept_)
print("R² Score (Sklearn):", model.score(x_reshaped, y))


# STEP 8: FINAL RESULT
print("\nRESULT:")
print("Thus, the linear regression model was implemented using both manual and sklearn methods successfully.")


print()
print()
print()
```

Choose Files headbrain.csv

headbrain.csv(text/csv) - 3370 bytes, last modified: 4/11/2026 - 100% done

```
Saving headbrain.csv to headbrain (1).csv
Dataset Preview:
    Gender  Age Range  Head Size(cm^3)  Brain Weight(grams)
0       1          1             4512                 1530
1       1          1             3738                 1297
2       1          1             4261                 1335
3       1          1             3777                 1282
4       1          1             4177                 1590

First 5 values of X: [4512 3738 4261 3777 4177]
First 5 values of Y: [1530 1297 1335 1282 1590]

Mean of X: 3633.9915611814345
Mean of Y: 1282.873417721519
Slope (m): 0.2634293394893993
Intercept (c): 325.5734210494428
```